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DEPARTMENT OF MASTERS OF COMPUTER APPLICATION
BALLARPUR INSTITUTE OF TECHNOLOGY, BAMNI, BALLARPUR



PROJECT OVERVIEW
ON
“SMART FARM MANAGEMENT APPLICATION”

Course :- MCA Semester II
2025 - 2026

HEAD OF DEPARTMENT:- **PROF. NEEHAL JIWANE**

GUIDED :- **PROF. INDU GADAPPA**

PRESENTATION BY:- **SABINA MOH SAFIK KHAN**

PRESENTATION BY:- **KANCHAN SANJAY BHUSHAN**



INTRODUCTION

- Agriculture is backbone of economy
- Traditional farming lacks digital management
- Need for smart and automated system
- 👉 “This project helps farmers manage data digitally”



PROBLEM STATEMENT

- Manual record keeping is difficult
- No proper income/expense tracking
- Lack of centralized system
- Data loss chances high



PROJECT OBJECTIVE

- Digitize farming records
- Manage farmers, crops, income & expenses
- Provide dashboard analytics
- Improve productivity using data



PROPOSED SYSTEM

- Desktop based application
- Built using Python Tkinter
- MySQL database for storage
- Easy and user-friendly interface



MODULES OF SYSTEM

- Login System
- Farmer Management
- Crop Management
- Expense Management
- Income Management
- Dashboard Analysis



TECHNOLOGY USED

- Python
- Tkinter (GUI)
- MySQL Database
- Pillow (Image handling)



SYSTEM ARCHITECTURE

- User → Tkinter UI → Python Logic → MySQL Database

SCREENSHOTS – LOGIN PAGE



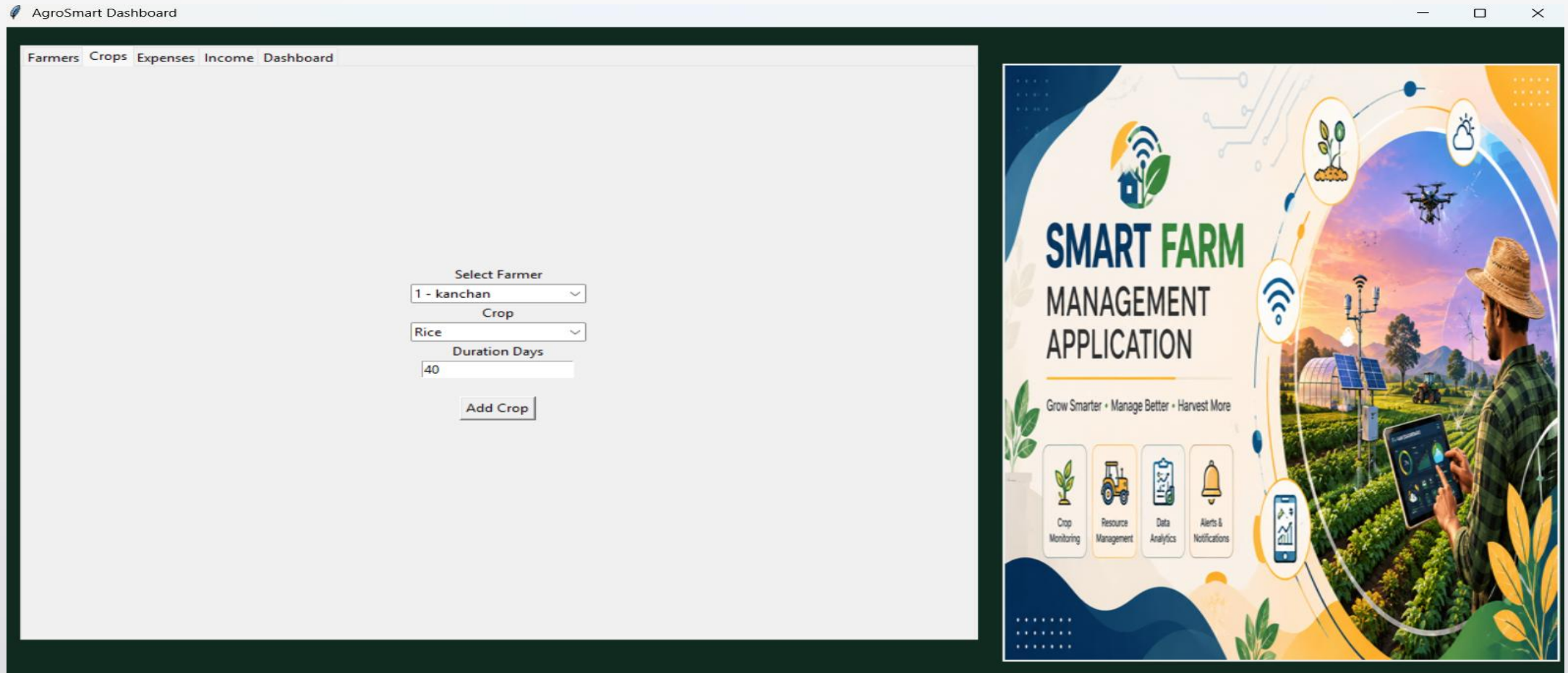
SCREENSHOT – FARMER MODULE

The screenshot displays the AgroSmart Dashboard interface. The top navigation bar includes links for Farmers, Crops, Expenses, Income, and Dashboard. The main content area is divided into two sections: a 'Farmer Form' and a 'Farmer List'. The 'Farmer Form' section contains a 'Farmer Details' form with the following fields:

- Name
- Phone
- Village
- Land Area (Acres)

Below the form fields are three buttons: 'Add' (green), 'Update' (grey), and 'Delete' (grey). To the right of the form is a large, colorful banner for the 'SMART FARM MANAGEMENT APPLICATION'. The banner features a farmer wearing a hat and holding a tablet, with a drone flying above a field. The text on the banner reads: 'SMART FARM MANAGEMENT APPLICATION', 'Grow Smarter • Manage Better • Harvest More', and 'Crop Monitoring', 'Resource Management', 'Data Analytics', 'Alerts & Notifications'.

SCREENSHOT- CROP MODULE



SCREENSHOT- EXPENSE & INCOME

AgroSmart Dashboard

Farmers Crops Expenses Income Dashboard

Select Farmer
2 - sabina

Amount
85000

Add Expense

SMART FARM
MANAGEMENT
APPLICATION

Grow Smarter • Manage Better • Harvest More

Crop Monitoring Resource Management Data Analytics Alerts & Notifications

AgroSmart Dashboard

Farmers Crops Expenses Income Dashboard

Select Farmer
1 - kanchan

Amount
750000

Add Income

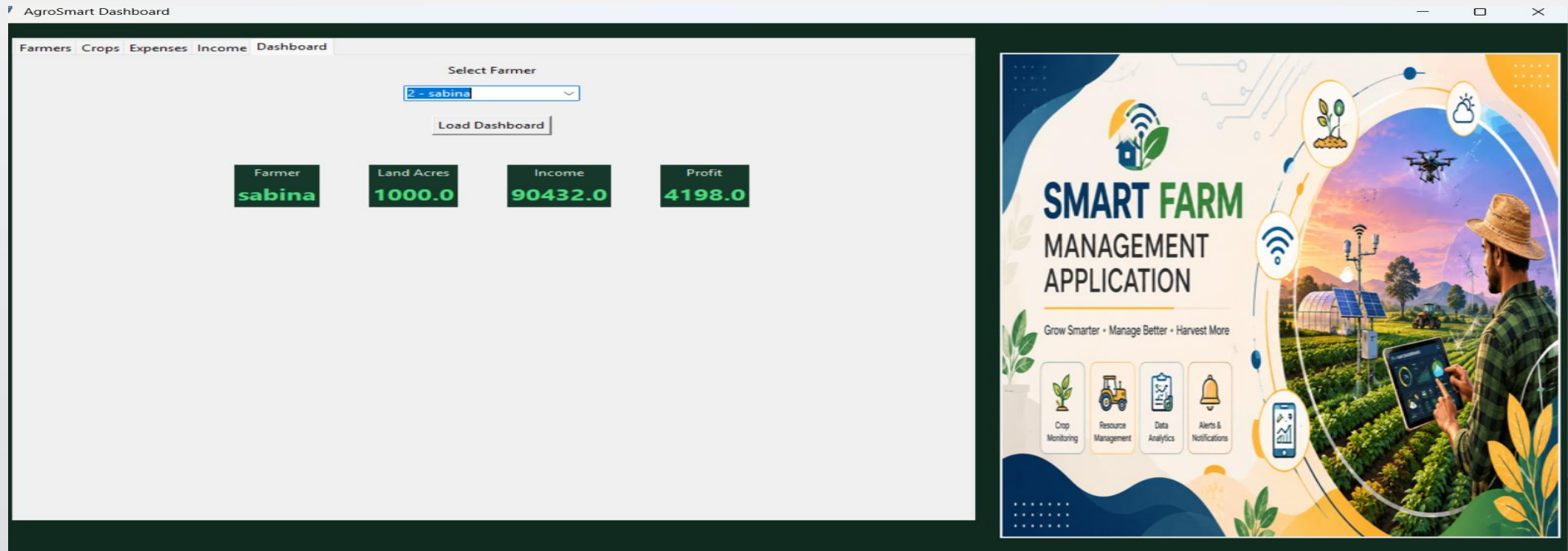
SMART FARM
MANAGEMENT
APPLICATION

Grow Smarter • Manage Better • Harvest More

Crop Monitoring Resource Management Data Analytics Alerts & Notifications

DASHBOARD

- Farmer, Land Acres, Income, Profit display
- Caption: “Real-time farm analytics dashboard”





FEATURES

- Easy GUI interface
- Data validation included
- Real-time calculations
- Secure login system
- Organized data storage



ADVANTAGES

- Saves time
- Reduces manual errors
- Improves productivity
- Easy data access



SCOPE

- Can be expanded to mobile app'
- Can include IoT sensors
- AI-based crop prediction possible



LIMITATIONS

- Desktop based only
- No online access
- No real-time weather API
- Limited scalability



CONCLUSION

- Smart solution for agriculture
- Helps in digital farming management
- Can be enhanced in future



THANK YOU